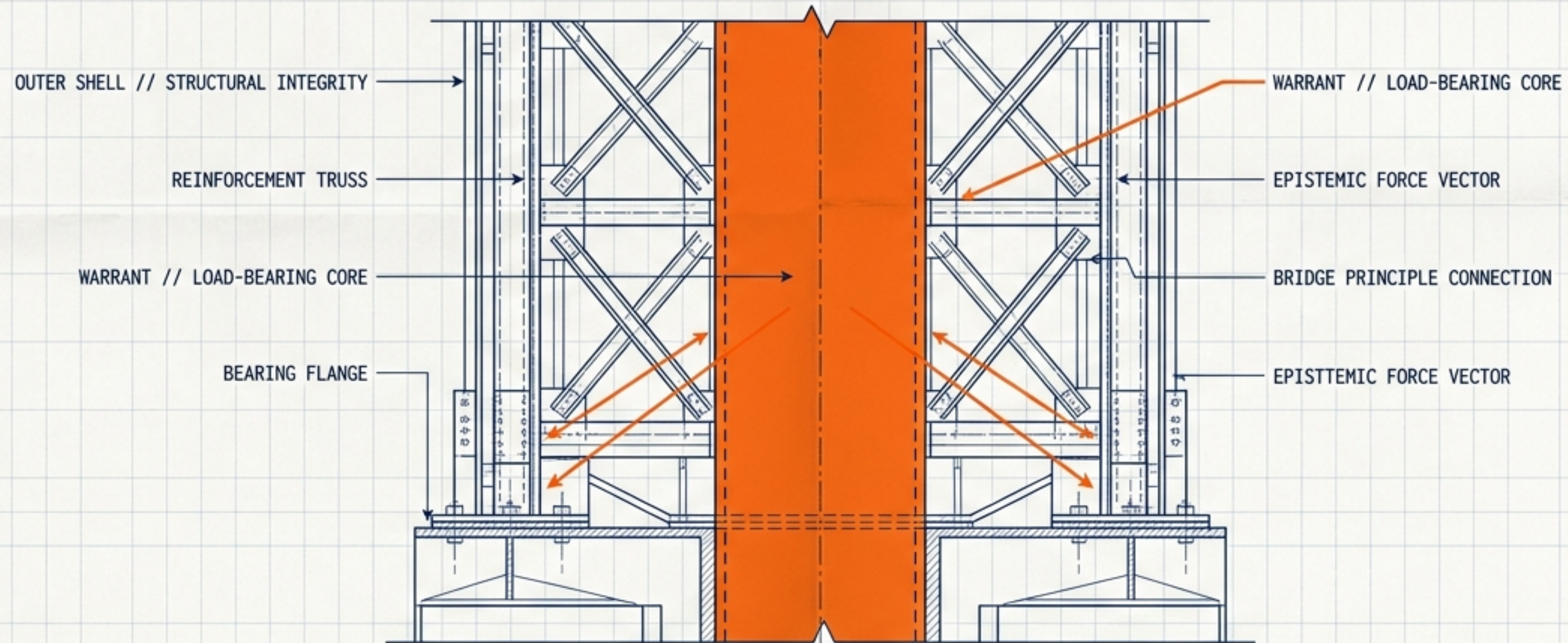


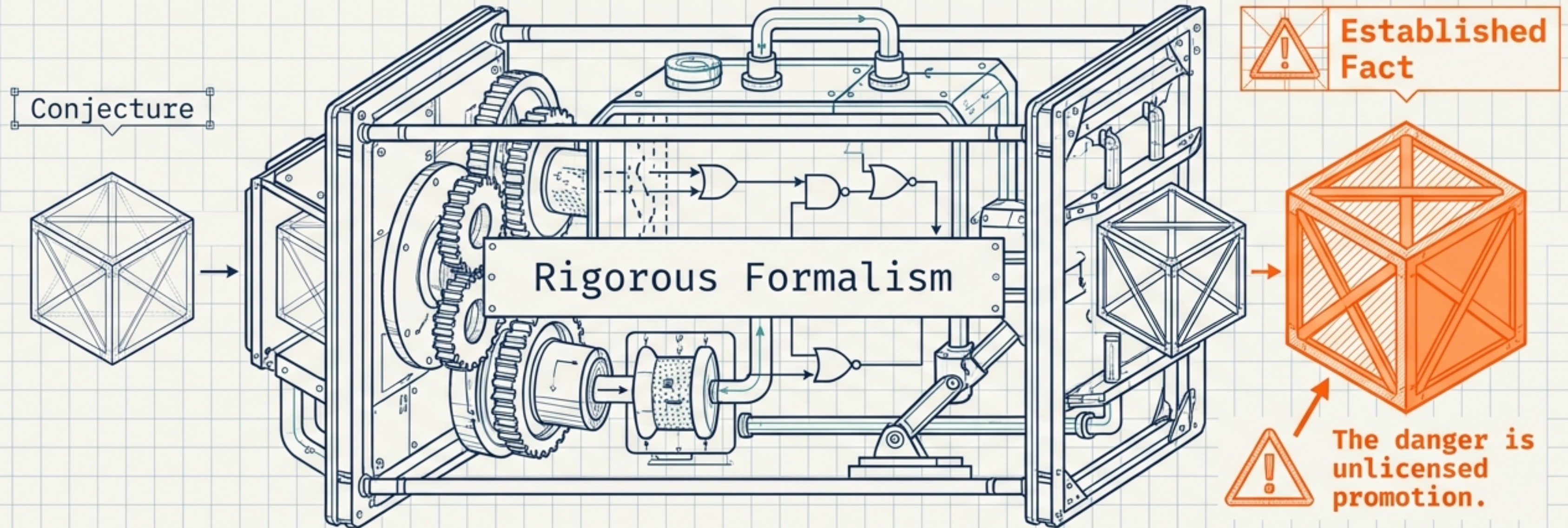
STRUCTURAL HONESTY

The Preservation of Recoverable Warrant Across Transformations.



INITIATING THE EPISTEMIC BLUEPRINT // SCALE: MACRO TO MICRO

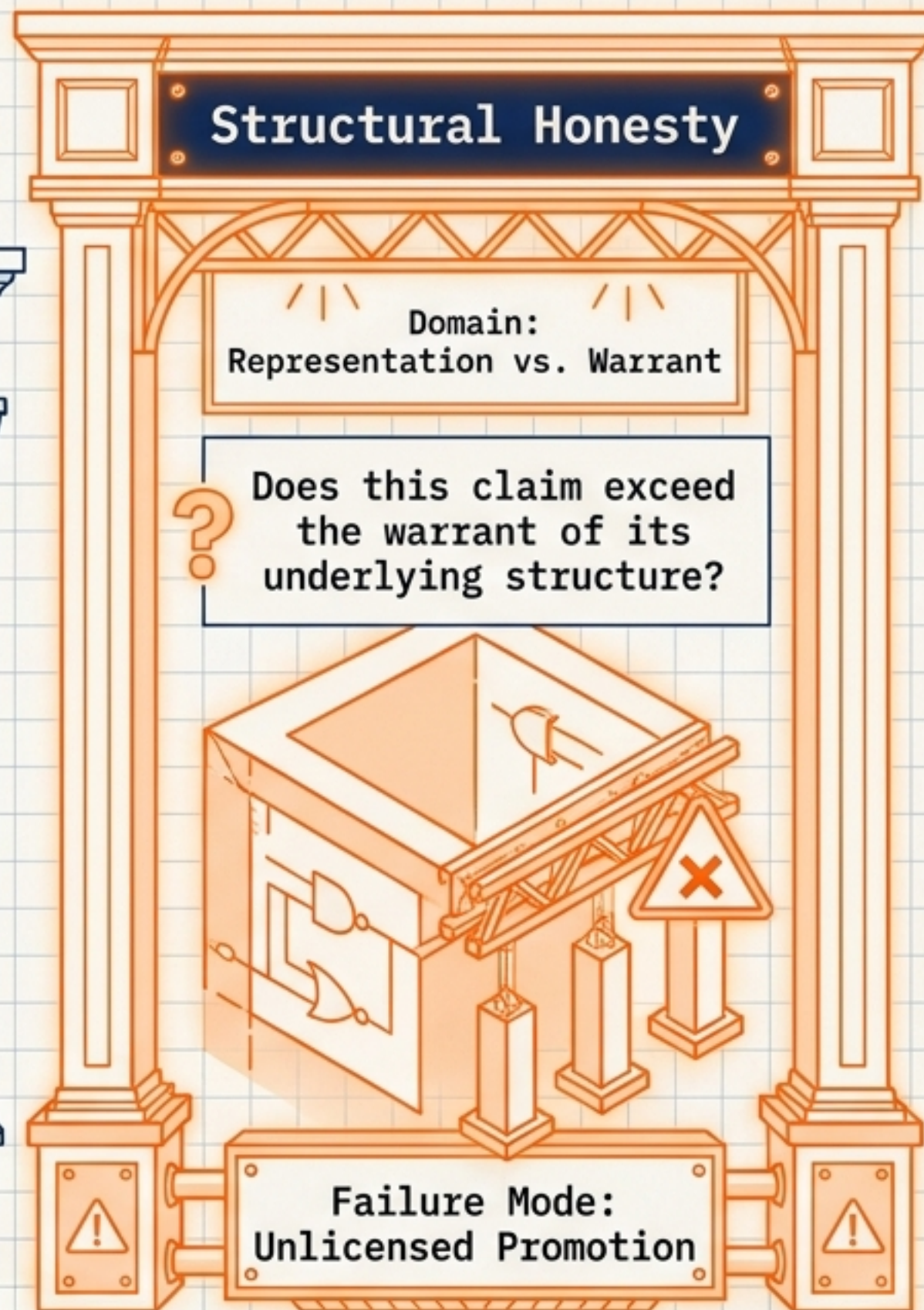
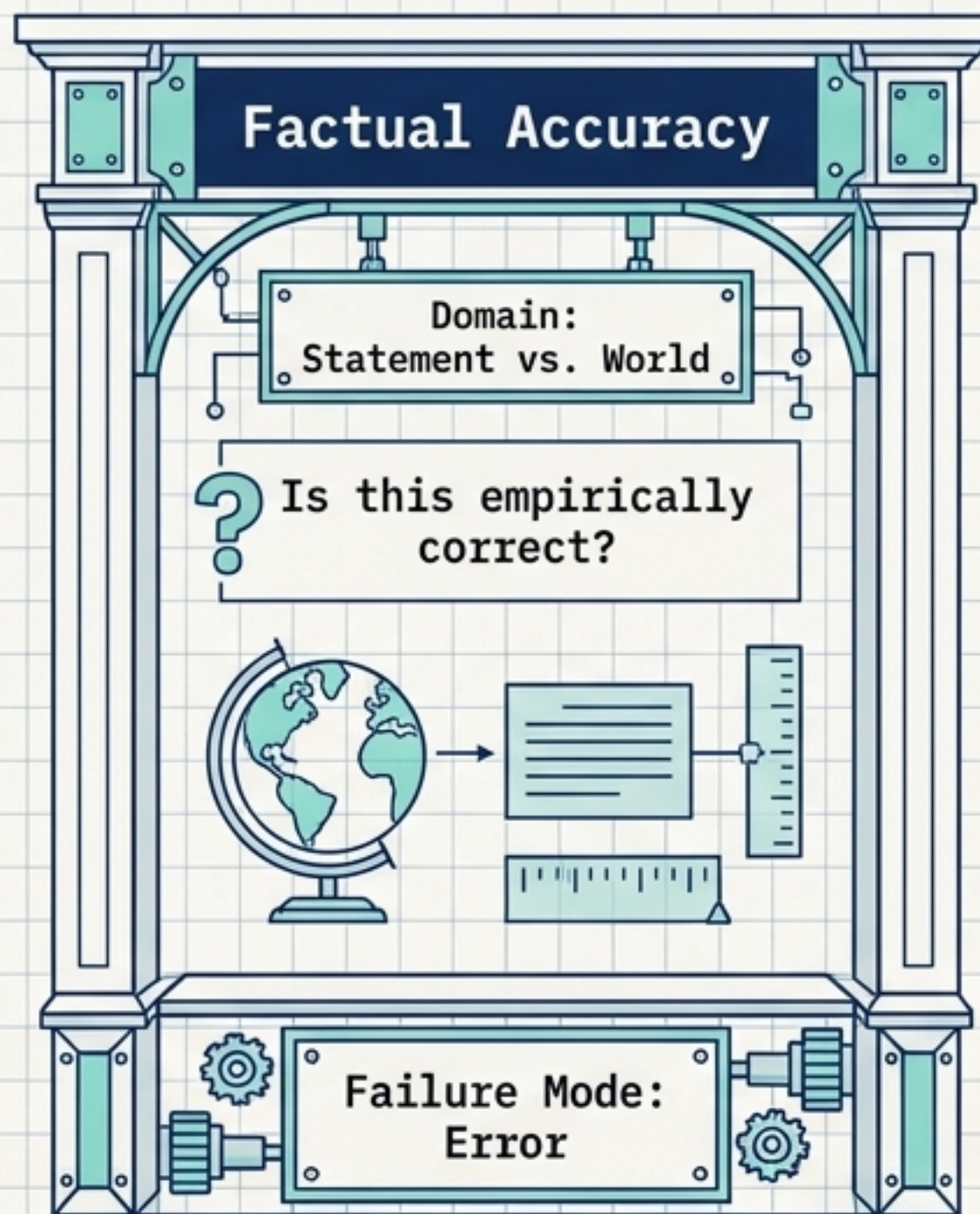
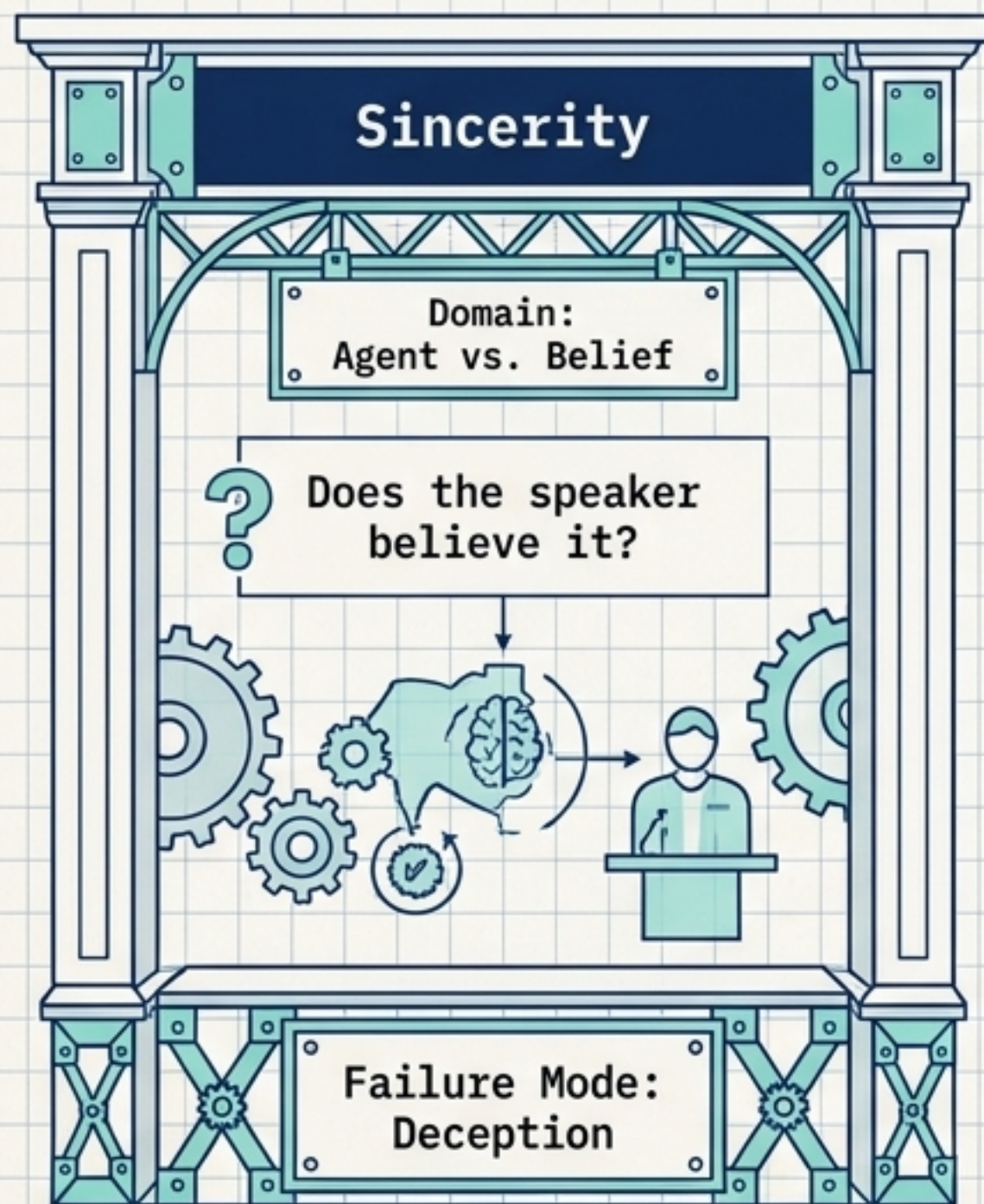
The Danger of Epistemic Laundering



A correct local operation, embedded in an otherwise rigorous structure, can quietly inherit a strength it was never separately shown to possess.

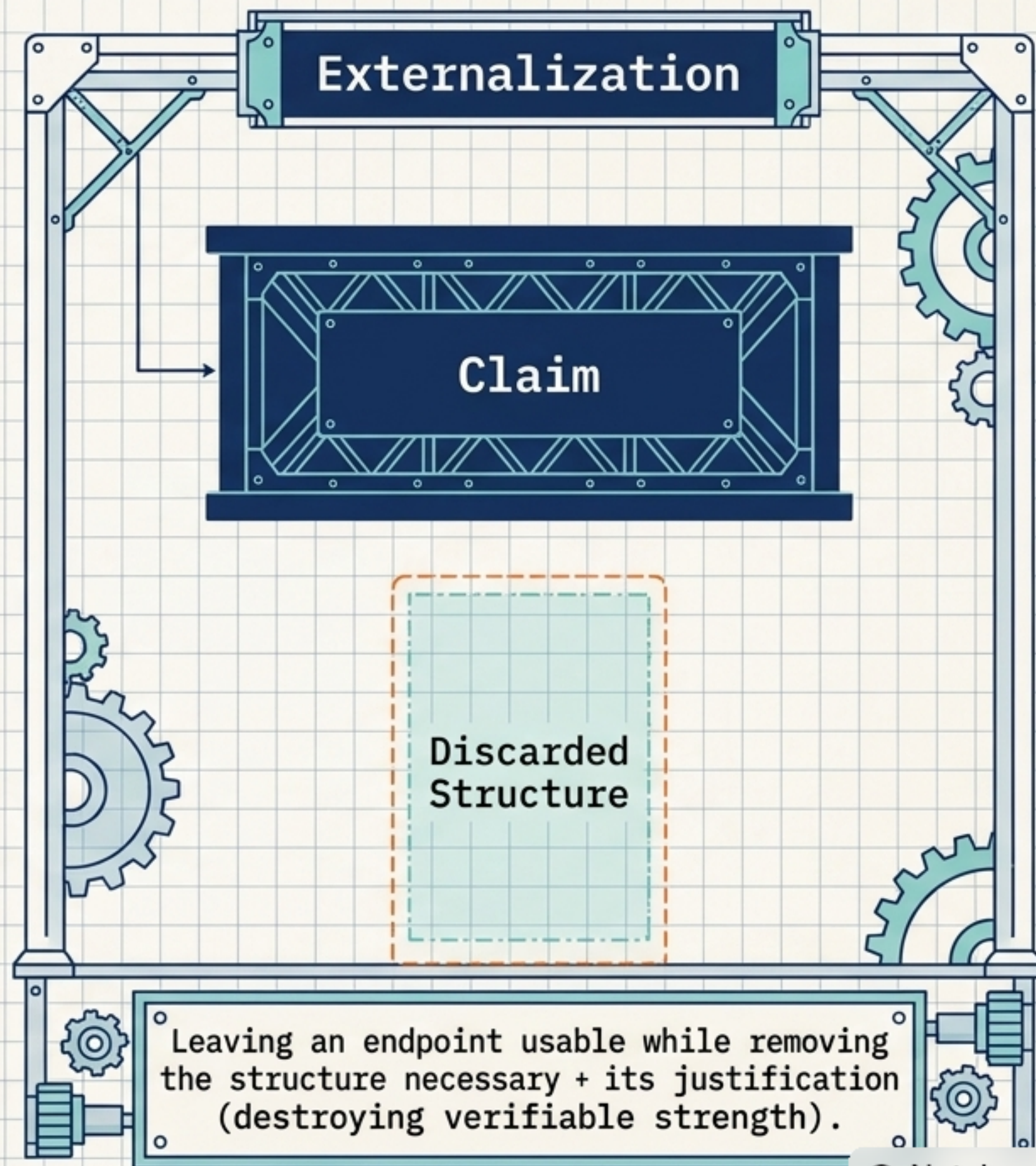
A numerical observation does not become an analytic result simply because the agreement is striking.
The danger is not ordinary falsehood. The danger is structural.

The Integrity Triad: A Structural Distinction

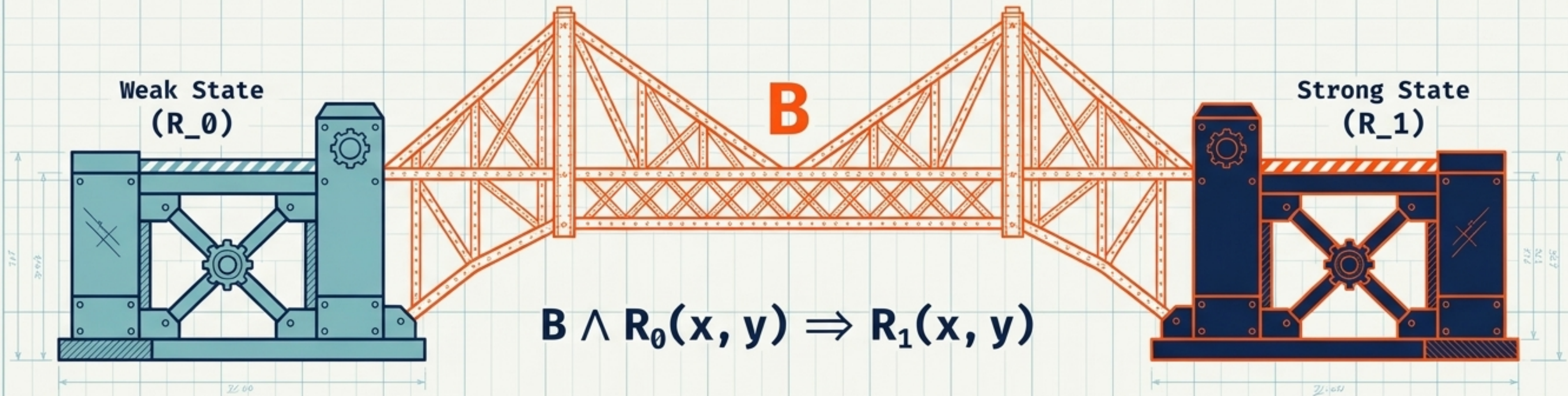


Takeaway: Structural dishonesty does not require deception, and it does not require that any individual statement be false.

The Two Modalities of Structural Failure



Bridge Principles & The Warrant Budget



The Warrant Budget Heuristic

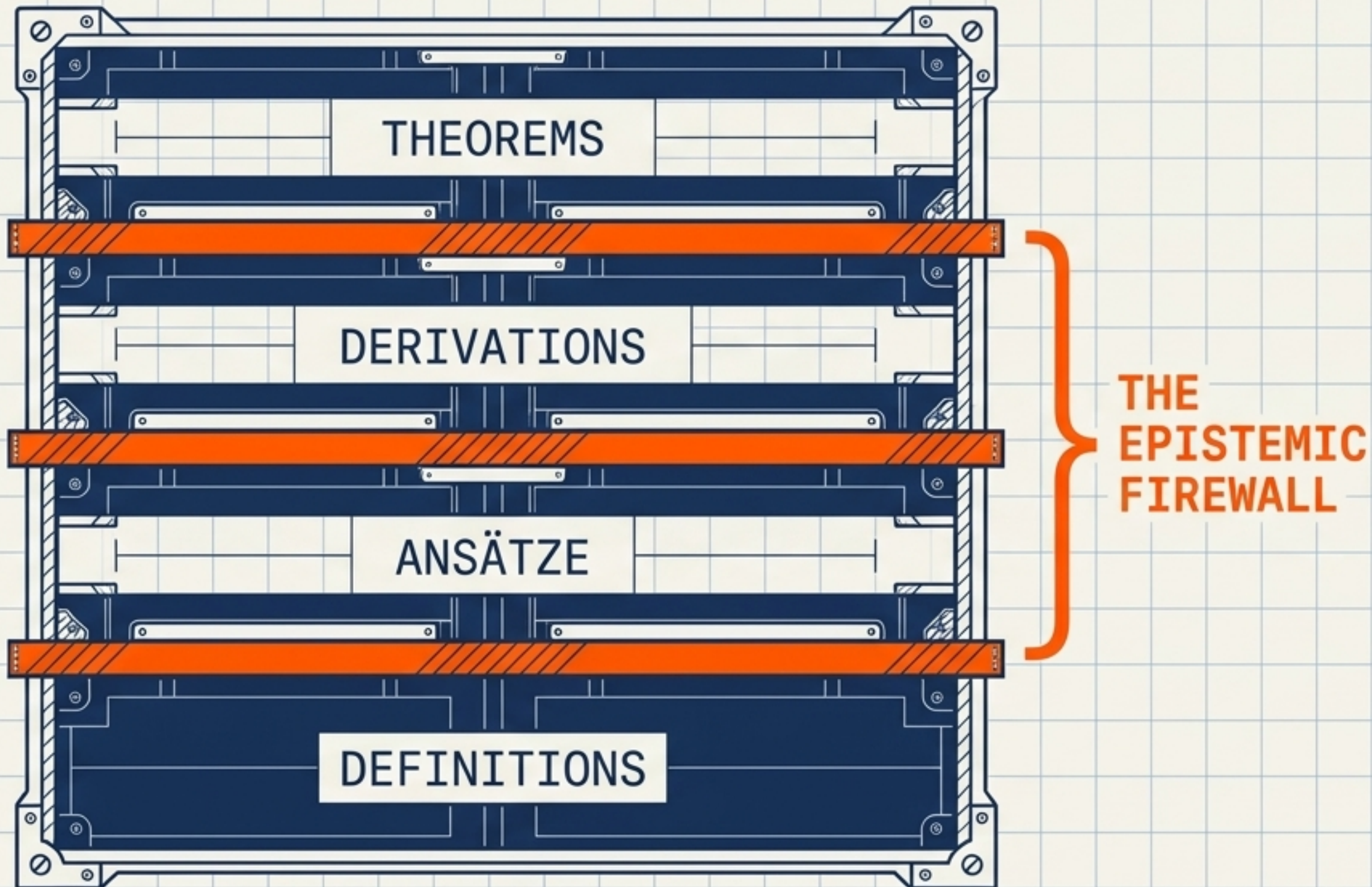
$$W(T_x) \leq W(x)$$

⚙️ A purely representational transformation (T) cannot increase epistemic entitlement simply by making an object more polished, compressed, or formal.

⚙️ If a conclusion appears stronger than its premises, the difference must have entered somewhere.

⚙️ **The Rule:** A weaker relation must not be silently promoted into a stronger one without an independent bridge principle.

Scale 1: Formalism & The Epistemic Firewall



Formalism must preserve the status of its premises.

- ✓ If $A \vdash P$, this establishes the inferential link, not the premise A .
- ✓ Downstream mathematical rigor cannot upgrade the epistemic status of an initial assumption.
- ✓ The Discipline: A structurally honest presentation keeps definitions, derivations, and conjectures in visibly distinct strata.

Scale 2: Enforcement in Code

CODE ARCHITECTURE

The Attestal Rust prototype implements **structural honesty** at the **type-system level**.

Result = Value + Generating History

A numerical result is irreversibly bound to its provenance record.

TERMINAL OUTPUT

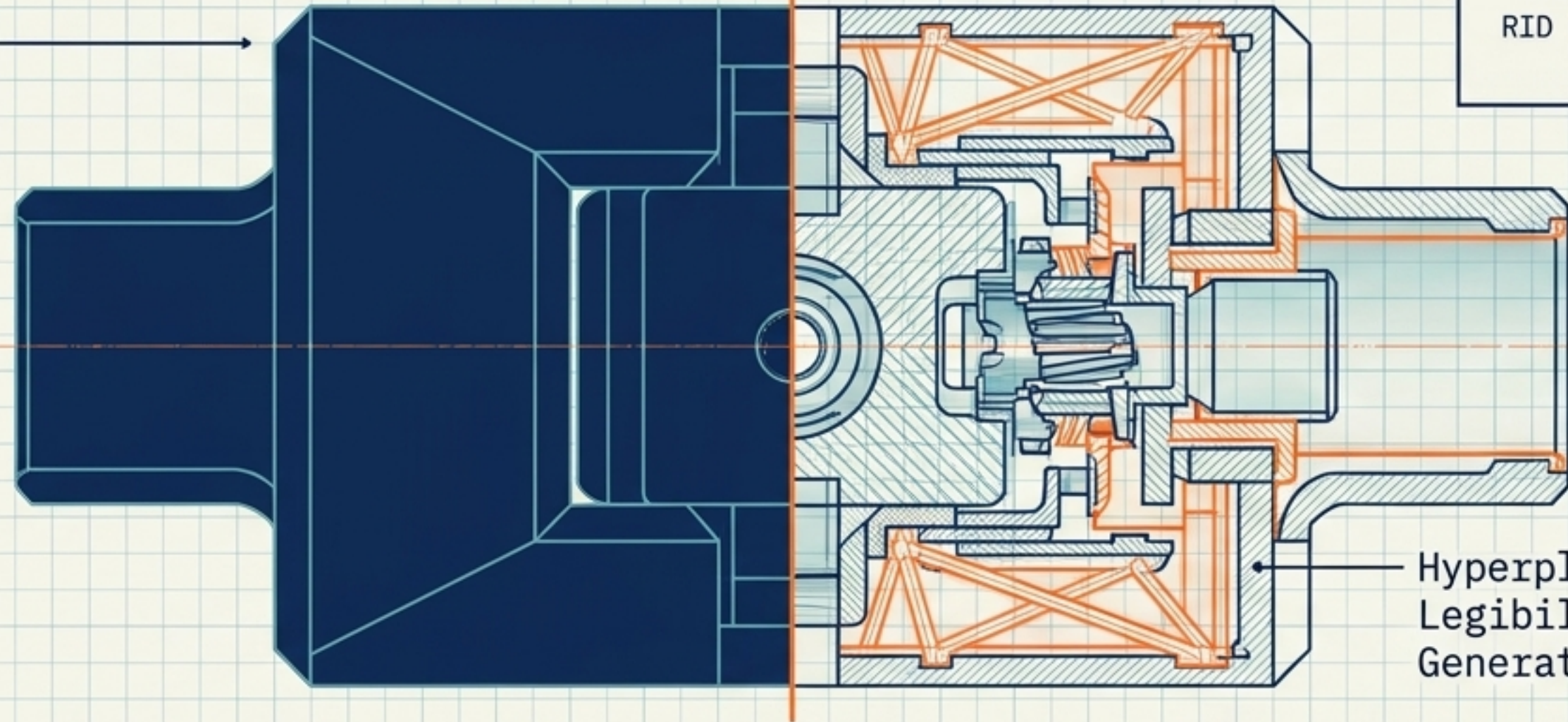
```
> require硬件()
```

```
[ERROR] claim rejected:  
attempted 'real hardware  
execution', but retained  
provenance says 'simulation  
(local prototype)'
```

Takeaway: The type system does not evaluate if the claim is plausible; it evaluates if the retained history licenses it.

Scale 3: Physical Artifacts & Generative Grammar

Externalized
Key ($K \notin A$)

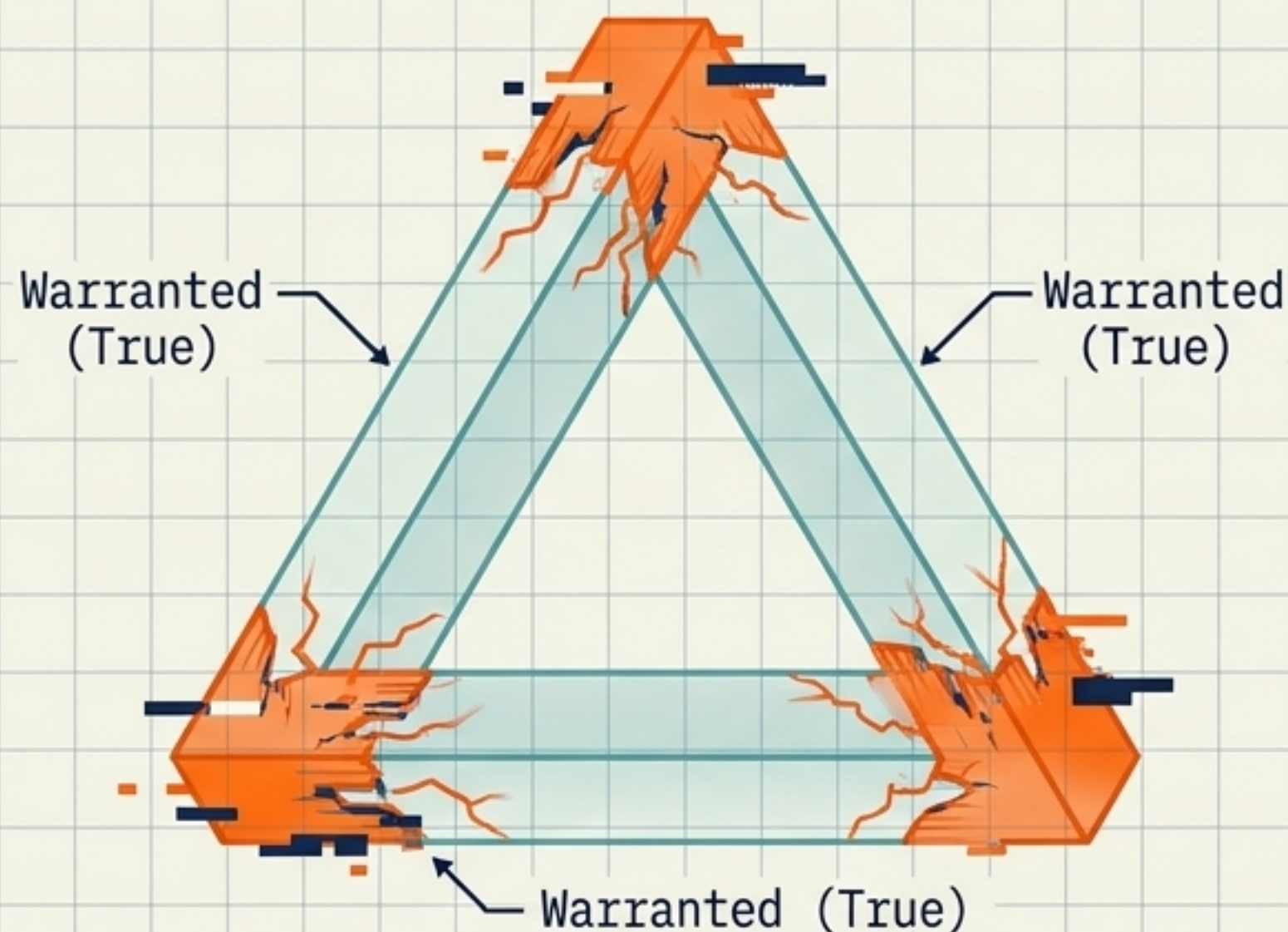


$$RID = \frac{I_{\text{recoverable}}}{I_{\text{total}}}$$

RID = Recoverable
Information Density

- An artifact A possesses a generative grammar G.
- In opaque systems, the interpretive key K is externalized. The object functions, but loses the capacity to explain itself.
- A structurally honest artifact preserves access to its generative status, keeping the conditions of interpretation attached to the object itself.

Scale 4: Compositional Dishonesty



$$\bigwedge_i W(c_i) \neq W(C(c_1, \dots, c_n))$$

Every component can be independently warranted, yet their composition can be structurally dishonest.

Ordering implies causation; juxtaposition implies comparison; shared scale implies commensurability.

None of these relations are guaranteed by the truth of the components.

The TARTAN Principle: Local consistency does not guarantee global realizability. Passing internal tests is not adequacy to the world.

Scale 5: Diagnostic Coordinates

Legibility Does Not Yet License Intervention.



Structural text

- Each transition in the diagnostic sequence requires its own independent justification (a bridge principle).

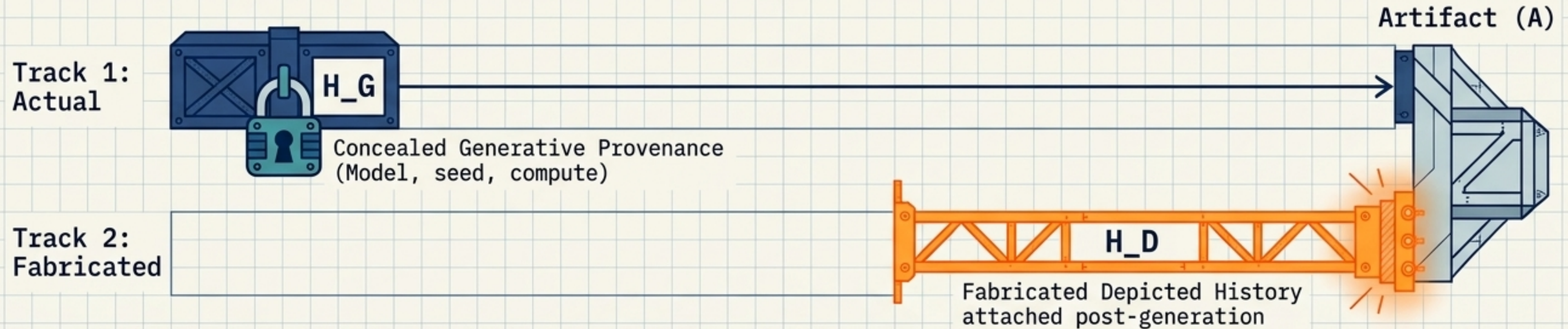
Structural text

- A detected deviation is evidence of a discrepancy; it is not a license to correct it.

Structural text

- The capacity to see what could be changed is a separate achievement from possessing the structural warrant to change it.
- The null intervention (doing nothing) must remain a live, visible option.

Scale 6: Synthetic Provenance & Time



$$\boxed{\text{Concealed Generative Provenance (H_G)}} + \boxed{\text{Fabricated Depicted History (H_D)}} = \boxed{\text{Provenance Substitution}}$$

Insight 1: Endpoint Limits

- **The Endpoint Cannot Testify:**
 $x(t) = y(t) \not\Rightarrow x \sim y$.
A shared present state does not imply a shared history.

Insight 2: Computational vs. Depicted

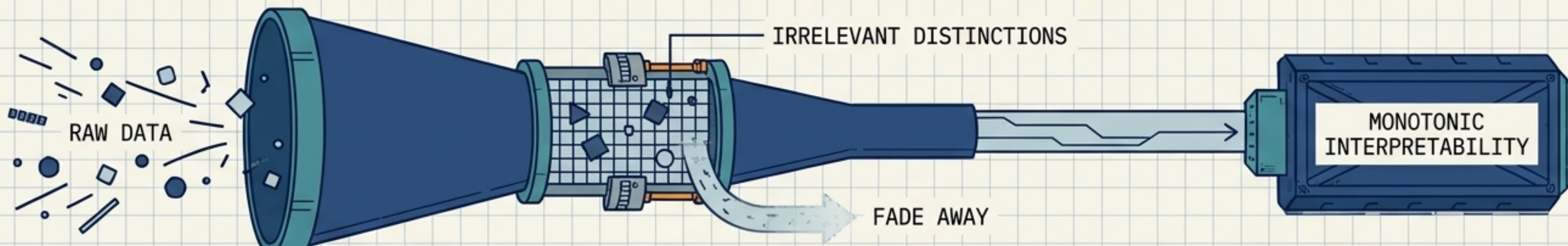
- Synthetic artifacts possess computational provenance (H_G) but are attached to fabricated depicted histories (H_D) they never experienced.

Insight 3: The Variant-Multiplicity Test

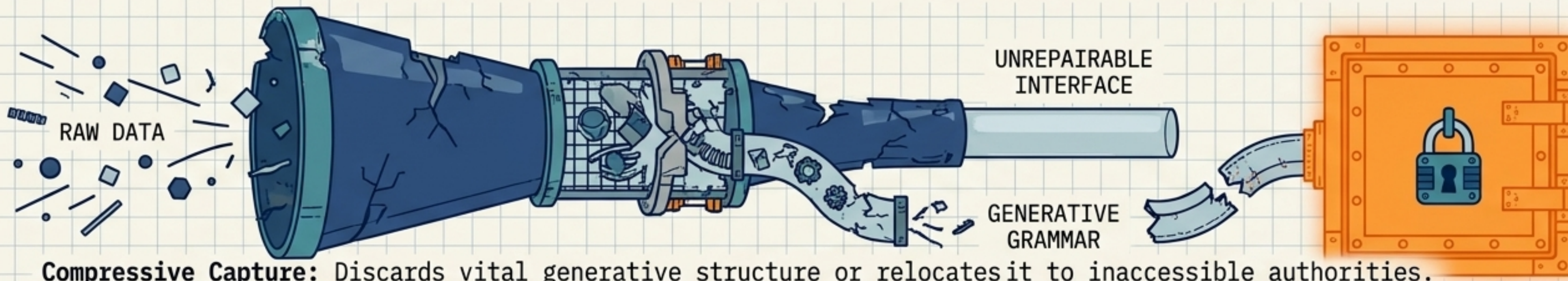
- When a 'unique' historical narrative has hundreds of mutually exclusive instantiations, it betrays a **generative template**, failing **global admissibility**.

The Right to Forget vs. Compressive Capture

An abstraction must earn the right to forget.



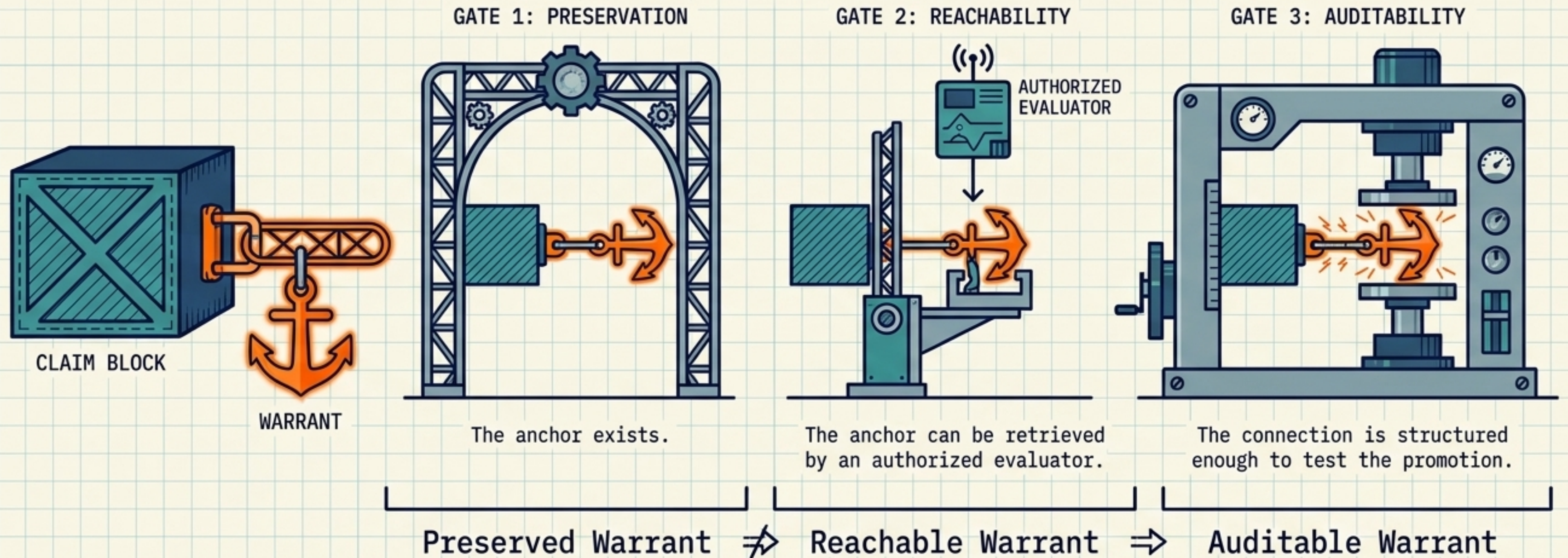
Honest Compression: Discards distinctions proven to be irrelevant. Allows monotonic refinement (experts can refine the coarse view without reversing it).



Compressive Capture: Discards vital generative structure or relocates it to inaccessible authorities.

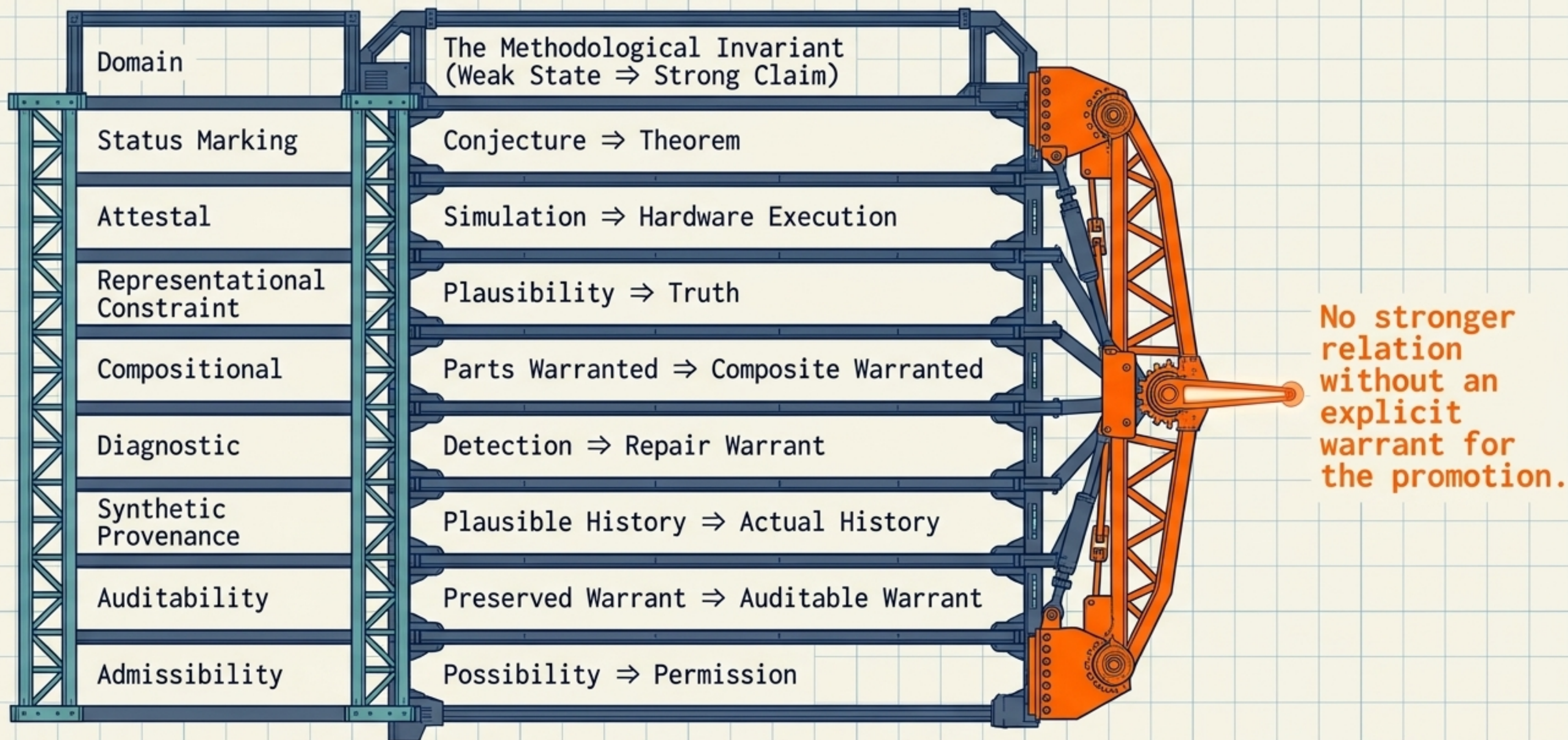
DIMENSIONAL POVERTY: Contraction of the control surface available to users for local repair. Interpretive power is redistributed, not just lost.

The Auditability Funnel: Warrant Must Survive Transport



A claim whose warrant predictably disappears on export, compression, quotation, or aggregation is only temporarily honest. Underdetermination must remain visible: "Unknown" is a positive epistemic state that prevents narrative substitution.

The Methodological Invariant



The Architecture of Structural Honesty

The Autonomy of Refusal

1. A claim must earn the strength of its assertion.

2. An abstraction must earn the right to forget.

3. An artifact must earn the confidence its function invites.

4. An artifact must not testify to a history it did not have.

5. Every increase in epistemic strength must have a traceable source.